

Background Literature Review: Urinary Phenol Metabolites

Is there an association in food levels and these concentrations

Method Section:

- **Sample Population starts at 2003-Present**
 - **Source Population: Inclusion & Exclusion Criteria (children):** Participants aged 6 years and older who met the subsample requirements.
 - Every individual who is eligible, sampling to see who had food insecurity data, phenol data, and coverient data
 - Among thodse, who had data that was above...
 - Propose to looking at strictly adults, adults are the ones responding especially since we are looking at households and
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 - **Urine specimens are collected, processed, and stored**
- NHANES Laborartory Data → Environmental Phenols Data Sheet
- List out study population: NHANES, what is NHANES?, questionnaire data on nutrition and health,

Limitations/Strengths:

- **Crossectional Study:** Can't establish temporality, snapshot, etc, urine is taken at one time point (phenolds are more stable than phthalates),
- **Strength:** Representative Study, urine sample collection (best scientifically),

Questions Slide:

- Come up with questions sbased off of my research on phthalates.

Literature Review

Agency for Toxic Substances and Disease Registry: Toxicological Profile for Phenol

- Commercial phenol is a liquid that ecaporates ore slowly than water
- Manufactured or produced naturally —> highly produced in the United States. It is used to make plastics, found in cleaning products, etc.
- Can persist in water for a week or more
- **Found in Food:** Low levels of phenol have been found in foods such as smoked summer sausage, smoked pork belly, mountain cheese, fried bacon, fried chicken, and black fermented tea.
 - Ingestion
- **Health Outcomes:**
 - Long-term exposure to phenol at work has been associated with cardiovascular disease, but the workers were also exposed to other chemicals at the same time.

Breast Cancer and the Environment Research Centers. (n.d.). *Early life exposure to phenols and breast cancer risk in later years: Fact sheet on phenols.*

https://bcerp.org/wp-content/uploads/2_BCERC.FactSheet_Phenols.pdf

- Phenols or phenolics are a class of organic chemical compounds that consists of one or more hydroxyl groups attached to an aromatic hydrocarbon group.
- They are either produced naturally or manufactured as a chemical.
- The main uses of phenols that are manufactured are appliance industries, automotive, adhesive, antiseptics, consumer products, etc.
- Humans can be exposed to phenols in multiple workplaces such as in the household, disinfectants, skin contact, contaminated water, or food.
- Free and bound phenol compounds are found naturally in foods. There are several foods that are known to have high concentrations of food: fruits (apples, bananas, oranges, red grapes, colored fruits, peanuts, tomatoes, milk etc. Furthermore, phenols can serve as a contaminant in packaged foods.
- **Bisphenol A (BPA)**
 - polycarbonate containers, cans, dental sealant, etc.
 - It is found in epoxy resins, which is used to line metal food and drink cans. Human exposure to BPA is mainly through ingestion.
 - Characterized that can act as a disruptor
- **Triclosan (TRCS)**
 - Microbicide in home cleaning and personal care products
- Some phenols can be endocrine disruptors, but it is likely that some phenolic compounds will not have any endocrine activity

Cao, X. L., Fu, W. H., & Popovic, S. (2024). Investigation of bisphenol S (BPS) in packaged fish, meat, cheese, and price labels on their corresponding packages. *Food additives & contaminants. Part A, Chemistry, analysis, control, exposure & risk assessment*, 41(11), 1509–1515. <https://doi.org/10.1080/19440049.2024.2394170>

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IUPAC Name	Other Name(s)	Common Name (Abbr.)	Main Uses	Food/Preservative Use	Key Chemical Properties
2,2-Bis(4-hydroxyphenyl)propane	4,4'-Isopropylidenediphenol	Bisphenol A (BPA)	Plastics, resins, can linings	Food-contact (linings)	Two phenols, weakly estrogenic, lipophilic
4,4'-Dihydroxydiphenylmethane	4,4'-Methylenediphenol	Bisphenol F (BPF)	Resins, BPA substitute	Food-contact (possible)	Two phenols, less bulky, endocrine active
4,4'-Sulfonyldiphenol	—	Bisphenol S (BPS)	BPA substitute, paper, plastics	Food-contact (possible)	Sulfone bridge, more water soluble, active
5-chloro-2-(2,4-dichlorophenoxy)phenol	—	Triclosan	Antibacterial in personal care, plastics	Food-contact (some articles, not additive)	Chlorinated ether, persistent, bioaccumulative
Methyl 4-hydroxybenzoate	p-Hydroxybenzoic acid methyl ester	Methylparaben	Preservative (cosmetics, pharma, food)	Food preservative (E218)	Paraben, water soluble, weakly estrogenic

Ethyl 4-hydroxybenzoate	p-Hydroxybenzoic acid ethyl ester	Ethylparaben	Preservative (cosmetics, pharma, food)	Food preservative (E214)	Paraben, sl. less soluble, estrogenic
Propyl 4-hydroxybenzoate	p-Hydroxybenzoic acid propyl ester	Propylparaben	Preservative (cosmetics, pharma, food)	Food preservative (E216)	Paraben, more lipophilic, estrogenic
Butyl 4-hydroxybenzoate	p-Hydroxybenzoic acid butyl ester	Butylparaben	Preservative (cosmetics, pharma; foods in some regions)	Food preservative (restricted)	Paraben, most lipophilic, estrogenic
2,4,5-Trichlorophenol	—	2,4,5-Trichlorophenol	Pesticide, wood preservative	No food use	Chlorinated, persistent
2,4-Dichlorophenol	—	2,4-Dichlorophenol	Herbicide precursor, disinfectant	No food use	Chlorinated, more acidic, pollutant
2,3,4,5,6-Pentachlorophenol	Pentachlorophenol, PCP	Pentachlorophenol (PCP)	Wood preservative, pesticide	No food use	Highly chlorinated, highly toxic, persistent
4-(2-propenyl)-2-methoxyphenol	Eugenol	Eugenol	Flavoring, dental anesthetic	Food flavoring	Allyl/methoxy, aromatic ether, volatile
5-methyl-2-isopropylphenol	2-Isopropyl-5-methylphenol, Thymol	Thymol	Antiseptic, flavoring, mouthwash	Food flavoring	Bulky, aromatic hydroxy, strong aroma
1,2-Dihydroxybenzene	Catechol	Catechol	Photography, flavoring, antioxidant	Trace in foods; rare flavoring	Ortho-dihydroxy, chelating, oxidizes easily

Analysis Plan:

Title: The Association between Household Food Security and Urinary Phenol Concentrations

Authors (*alphabetical order*): Ayishat Yussuf, Marshae Nickelberry, and Tamarra James-Todd.

Purpose: To examine the association between household food security and urinary phenol concentrations amongst the national U.S. population, using the NHANES database (2003-2016).

Background

Food security is defined as "the reliable access to enough food for a healthy lifestyle" (USDA, 2025), which food insecurity refers to limited or uncertain access due to financial or resource constraints. Approximately 17.9 million households in the U.S. were considered food insecure in 2023 (USDA). Roughly 12.2 million adults lived in households with very low food security. Household food insecurity is not experienced equally across populations. Rates are consistently higher among Non-Hispanic Black and Hispanic households compared to Non-Hispanic White households. Food insecurity often leads to changes in both the quantity and quality of food consumed, with households facing limited access to fresh, whole foods and relying more heavily on packaged, processed, and shelf-stable items.

Phenols are a class of organic chemical compounds, characterized by one or more hydroxyl groups attached to an aromatic hydrocarbon ring. These chemicals are widely used in the manufacturing of common household items, food packaging, and food preservatives. As a result, humans are exposed to phenols through multiple pathways, including inhalation, ingestion, and skin contact.

One key dietary pathway of phenol exposure is food packaging and storage. Phenols such as bisphenol A (BPA) are used in epoxy resins and polycarbonate plastics, which are common in canned foods, plastic containers, and bottled beverages. BPA and related compounds can leach into food and drinks, particularly when packaging is exposed to heat, processing, or long-term storage. Ingestion therefore represents the primary source of human exposure to these chemicals.

Health outcomes linked to phenol exposure include endocrine disruption, cardiovascular effects, and certain cancers. Previous literature has examined the connection between phenol exposure and diet, showing that consumption of ultra-processed foods is associated with higher urinary concentrations of several phenols. For example, Buckley et al. (2020) reported that certain ultra-processed food groups were associated with higher urinary bisphenol A (soft and fruit drinks), bisphenol F (French fries, potato products, milk-based desserts), and bisphenol S (pizza), while other foods were linked to lower bisphenol concentrations.

Despite these findings, a key research gap remains: how exposures to phenols, especially through dietary pathways, may be linked to household food insecurity. Understanding this connection is important for identifying environmental health disparities and the broader health impacts of food insecurity.

Hypotheses:

(1) The hypothesis of this project is that increasing levels of food insecurity will be linked to higher exposure to environmental contaminants.

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(2) Race/ethnicity and sex will modify the association.

Study population

- **Study population will include:**
 - Participants in the NHANES research program with laboratory data and with available data for food security, demographic information, and phenol concentrations.
- **Study population will exclude:**
 - Participants who have missing data for the food security section
 - Participants missing demographic information (gender, ethnicity, age, education)
 - Participants who are missing data for phenol concentrations

Exposure

- **USDA Household Food Security Survey Module (FSSM) (Ordinal Variable)**
- Based on a well-validated 18-item module that measures the severity of household food security.
 - 1 – fully food secure (0) (reference group)
 - 2 – marginally food secure (1-2)
 - 3 – low food security (3-5 for HH without child, 3-7 for HH with child)
 - 4 – very low food security (6-10 for HH without child, 8-18 for HH with child)

Code or Value	Value Description	Definition
1	HH fully food secure	Households show no or minimal evidence of food insecurity.
2	HH marginally food secure	Food insecurity is evident in household members' concerns about adequacy of the household food supply and in adjustments to household food management, including reduced quality of food and increased unusual coping patterns. Little or no reduction in members' food intake is reported.
3	HH food insecure without hunger	Food insecurity is evident in household members' concerns about adequacy of the household food supply and in adjustments to household food management, including reduced quality of food and increased unusual coping patterns. Little or no reduction in members' food intake is reported.
4	HH food insecure with hunger	At this level, all households with children have reduced the children's food intake to an extent indicating that the children have experienced hunger. For some other households with children, this already has occurred at an earlier stage of severity. Adults in households with and without children have repeatedly experienced more extensive reductions in food intake.

Outcome

Phenol urinary concentration (Continuous Variable)

- Phenols are measured in ng/mL.
- Measured by spot urine samples, assayed by the CDC using standardized techniques
- Adjusted for creatinine for urinary dilution correction

Two variables are provided for each of the analytes:

- Each phenol has two variables provided by NHANES: One continuous variable with the prefix “URX” and a detection flag with the prefix “URD” ending in LC, indicating whether the value was below the lower limit of detection (LLOD).
 - Value of 0 = the concentration reported is above the LLOD
 - Value of 1 = the concentration reported is below LLOD
- Any values below the LLOD were substituted with $LLOD/\sqrt{2}$.

There are numerous phenols that NHANES has measured throughout several data cycles, but not all phenols were measured each year. We decided to analyze the following **9 phenolic metabolites**:

1. Urinary Bisphenol A - URXBPH
2. Urinary Bisphenol F - URXBPF
3. Urinary Bisphenol S - URXBPS
4. Urinary 2-Hydroxy-4-methoxybenzophenone (Benzophenone-3) - URXBP3
5. Urinary 2,4,4'-Trichloro-2'-hydroxyphenyl ether (Triclosan) - URXTRS
6. Urinary Butyl paraben - URXBUP
7. Urinary Propyl paraben - URXPPB
8. Urinary Ethyl paraben - URXEPB
9. Urinary Methyl paraben - URXMPB

Confounders: Referenced 2015-2016 Data Documentation

Age

- **Variable Name:** RIDAGEYR - Age in years at screening
- **Variable Type:** Ordinal
- **Code/Values:**
 - **0 to 79** - Range of Values
 - **80** - 80 years of age and over
 - 1 = <20 years
 - 2 = 20 – 29 years
 - 3 = 30 - 39 years
 - 4 = 40 – 49 years
 - 5 = 50 – 59 years
 - 6 = 60 – 69 years
 - 7 = 70 – 79 years
 - 8 = ≥80 years

Race/Ethnicity:

- **Variable Name:** RIDRETH1
- **Variable Type:** Nominal
- **Code/Values:**
 - 1 = Mexican American
 - 2 = Other Hispanic
 - 3 = Non-Hispanic White
 - 4 = Non-Hispanic Black
 - 5 = Other Race - Including Multi-Racial

Education

- **Variable Names:**
 - Children Youth 6-19 - DMDEDUC3
 - Adults 20+ years: DMDEDUC2
- **Variable Type:** Ordinal

Recode: Educat

- **Code/Values:**
 - 0 = Never attended/Kindergarten Only
 - 1 = 'less than high school'
 - 2 = 'high school/ GED equivalent'
 - 3 = 'more than high school'

Sex:

- **Variable Name:** RIAGENDR
- **Variable Type:** Nominal
- **Code/Values:**
 - 1 - Male
 - 2 - Female

Family income to poverty ratio:

- Measured by The Department of Health Human Services (HHS) poverty guidelines, utilized to calculate the ratio.

Variable Name: INDFMPIR

Variable Type: Continuous

0 to 4.99 = Range of Values

5 = Value greater than or equal to 5.00

Survey Cycle Years

Variable Name: INDFMPIR

Variable Type: Nominal

- **Code/Values:**
 - 2003/2004 – 1
 - 2005/2006 – 2
 - 2007/2008 – 3
 - 2009/2010 – 4
 - 2011/2012 – 5

- 2013/2014 – 6
- 2015/2016 – 7

Creatinine

Variable Name: URXUCR

Variable Type: (Continuous/Scale Variable)

Statistical Analysis

Descriptive : Frequencies of the covariate variables by food security status

Bivariate: Chi-square test on categorical variables, ANOVA on continuous variables

Multivariate: Multivariate linear regression models

- Adjusted percent differences in phenol concentrations.
- **Adjusted:** *Age (categorical), sex (binary), education (categorical), poverty: income (categorical), race/ethnicity (categorical), survey year(categorical), creatinine (continuous)*

Secondary: Effect measure modification w/ interaction terms

- Sex * Household Food Security
- Race/ethnicity * Household Food Security

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Poster Presentation Outline

Background:

Food secure is defined as a household having access to enough food to maintain a healthy lifestyle.

Food insecure is defined as a household that has faced times of not being able to acquire enough food due to difficult access to food, resources or the inability to afford produce. This includes low and very low food security.

According to the FDA, roughly 18 million households in the U.S. were considered food insecure in 2023.

6.5 million households with children under 18 years old.

12.2 million adults lived in household with very low food security.

- 841,000 children lived in very low food security households.

Methods Outline:

- Sample initially included 18,961 participants which includes adults and children but we removed children, which made the sample 13,207 (adults ≥ 18 years)
- Checked for missingness for participants who are missing covariate and exposure data, resulting in 10,423 participants.
 - **Covariates included:** Education, race/ethnicity, age, poverty income ratio, and cycle years
 - **Exposure:** Household Food security levels(ranging from levels 1-4)
 - 1 – fully food secure
 - 2 – marginally food secure
 - 3 – low food security
 - 4 – very low food security
 - **Outcome:** Phenol Concentrations
- Initially, we had the following phenols listed to examine:
 - **Urinary Bisphenol A - URXBPH**
 - Urinary Bisphenol F - URXBPF
 - Urinary Bisphenol S - URXBPS
 - **Urinary 2-Hydroxy-4-metoxybenzophenone (Benzophenone-3) - URXBP3**
 - **Urinary 2,4,4'-Trichloro-2'-hydroxyphenyl ether (Triclosan) - URXTRS**
 - **Urinary Butyl paraben - URXBUP**
 - **Urinary Propyl paraben - URXPPB**
 - **Urinary Ethyl paraben - URXEPP**
 - **Urinary Methyl paraben - URXMPB**

- After realizing that 2003-2004 only included three of the phenols (BPA, TRS, and BP3), we decided to start from 2005-2016 and only included the phenols highlighted and bolded, making this our final set of exposures (7 in total)
 - *BPF and BPS had excessive missingness and limited measurement years*
- In regard to the detection limit, any values below the LLOD were substituted with $LLOD/\sqrt{2}$, which is consistent with the NHANES analytic guidance within the laboratory results. Outliers greater than 5 standard deviations above the mean were excluded → *we had thoughts of altering this restriction.*
 - The CDC's analytic measures changed overtime so Marshae pointed out that TRS for example changes
- Furthermore, log transformations were coded for each phenol and histograms were created, which were all right skewed and the log-transformed to estimated/approximate normality based off of the NHANES data used..
- The subsample weights were adjusted for by dividing each year by the number of cycle years (n=7). The analyses included survey design variables (SDMVPSU and SDMVSTRA) and the appropriate weight based on phenols.
- After removing participants with missing exposure and outcome data, applying restrictions, the weighted analytic sample included roughly 9,700 participants.

Abstract:

Food security is defined as "the reliable access to enough food for a healthy lifestyle" (USDA, 2025), while food insecurity refers to limited or uncertain access due to financial or resource constraints. Previous literature has examined the link between phenol exposure and diet, showing that consumption of ultra-processed foods is associated with higher urinary concentrations of several phenols. The objective of this research project was to investigate the association between household food security and urinary phenol concentrations in the U.S. Individuals with household food insecurity had higher levels of certain urinary phenol metabolites that come from parent compounds used in food packaging and processed foods. Results revealed that households that are food insecure (Groups 3 and 4) have higher concentrations of... Future studies should investigate and evaluate the role of structural racism as a contributing factor of household and neighborhood food security and environmental chemical exposures.